Canonical Correlation Analysis

Readings: Johnson & Wichern, Chapter 10;
Zelterman, Section 13.2; Izenman, Section 7.3

Whitney Huang

Clemson University | DSA 8070

Agenda

Background

Canonical Variates & Canonical Correlations

Sales Data Example

Learning Objectives

By the end of this lecture, you should be able to:

- Explain why CCA is useful for relating **two sets** of variables.
- Construct and interpret canonical variate pairs.
- Explain the optimization and eigenvalue formulations.
- Use canonical correlations, significance tests, and loadings to interpret a CCA.
- Translate statistical findings into an applied conclusion.

A Business Question: What Relates to Sales Performance?

A company collects information from $n = 50$ employees.

Sales performance ($p = 3$)

Sales growth; profitability; new-account sales

Assessment scores ($q = 4$)

Creativity; mechanical reasoning; abstract reasoning; mathematics

Question: How are these *two sets* of outcomes related? Pairwise analysis would involve $3 \times 4 = 12$ correlations. Can we summarize their shared patterns more effectively?

The Central Idea of Canonical Correlation Analysis (CCA)

Let $\mathbf{X} \in \mathbb{R}^p$ and $\mathbf{Y} \in \mathbb{R}^q$ represent two sets of measurements. Find linear combinations

$$U_1 = \mathbf{a}_1^\top \mathbf{X}, \qquad V_1 = \mathbf{b}_1^\top \mathbf{Y}$$

that have the **largest possible correlation**.

Interpretation

U_1 is a composite of the X variables; V_1 is a composite of the Y variables. CCA finds the most strongly associated pair of composites.

CCA first identifies the strongest association between the two sets, then looks for additional, uncorrelated pairs of linear combinations.

Historical Note: Origins of CCA

Introduced by Harold Hotelling (1936), CCA extends the idea of correlation from two variables to two sets of variables.

RELATIONS BETWEEN TWO SETS OF VARIATES*.

By HAROLD HOTELLING, Columbia University.

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1. *The Correlation of Vectors. The Most Predictable Criterion and the Tetrad Difference.* Concepts of correlation and regression may be applied not only to ordinary one-dimensional variates but also to variates of two or more dimensions.

CCA Compared with PCA

	PCA	CCA
Input	One set of variables	Two sets of variables
Objective	Maximize variance	Maximize correlation between paired composites
Output	Principal components	Pairs of canonical variates
Use	Dimension reduction within a set	Summarize association across sets

Key distinction: A direction with high variance need not be the direction most associated with another set.

Covariance Structure

Assume centered random vectors (i.e., each variable has mean zero) $\mathbf{X} \in \mathbb{R}^p$ and $\mathbf{Y} \in \mathbb{R}^q$, with

$$\operatorname{Var}\begin{pmatrix}\mathbf{X}\\ \mathbf{Y}\end{pmatrix} = \begin{pmatrix}\Sigma_{XX} & \Sigma_{XY}\\ \Sigma_{YX} & \Sigma_{YY}\end{pmatrix}.$$

For $U = \mathbf{a}^\top \mathbf{X}$ and $V = \mathbf{b}^\top \mathbf{Y}$,

$$\begin{aligned}\operatorname{Var}(U) &= \mathbf{a}^\top \Sigma_{XX} \mathbf{a},\\ \operatorname{Var}(V) &= \mathbf{b}^\top \Sigma_{YY} \mathbf{b},\\ \operatorname{Cov}(U, V) &= \mathbf{a}^\top \Sigma_{XY} \mathbf{b}.\end{aligned}$$

The **cross-covariance block** Σ_{XY} connects the two sets.

First Canonical Correlation: Optimization

$$\rho_1 = \max_{\mathbf{a},\mathbf{b}} \frac{\mathbf{a}^\top \Sigma_{XY} \mathbf{b}}{\sqrt{\mathbf{a}^\top \Sigma_{XX} \mathbf{a}} \sqrt{\mathbf{b}^\top \Sigma_{YY} \mathbf{b}}}.$$

Equivalently, maximize $\mathbf{a}^\top \Sigma_{XY} \mathbf{b}$ subject to

$$\mathbf{a}^\top \Sigma_{XX} \mathbf{a} = 1, \qquad \mathbf{b}^\top \Sigma_{YY} \mathbf{b} = 1.$$

Why constrain the variances?

Without normalization, multiplying coefficients by arbitrary constants changes the scale of the composites without changing their correlation.

Subsequent Canonical Variate Pairs

The second pair (U_2, V_2) maximizes correlation after removing the first pair's directions. Each canonical variate is normalized to

variance one, and

$$\mathrm{Cov}(U_i, U_j) = \mathrm{Cov}(V_i, V_j) = 0 \quad (i \neq j).$$

Canonical variates from *different* pairs also have zero cross-covariance:

$$\mathrm{Cov}(U_i, V_j) = 0 \quad (i \neq j).$$

Result: At most $\min(p, q)$ canonical pairs, ordered as

$$1 \geq \rho_1 \geq \rho_2 \geq \cdots \geq 0.$$

How Are the Directions Computed?

Assuming Σ_{XX} and Σ_{YY} are positive definite, define

$$C = \Sigma_{XX}^{-1/2} \Sigma_{XY} \Sigma_{YY}^{-1/2}.$$

The singular values of C are the canonical correlations:

$$C\mathbf{v}_i = \rho_i \mathbf{u}_i, \qquad C^\top \mathbf{u}_i = \rho_i \mathbf{v}_i.$$

Then the canonical coefficients are

$$\mathbf{a}_i = \Sigma_{XX}^{-1/2} \mathbf{u}_i, \qquad \mathbf{b}_i = \Sigma_{YY}^{-1/2} \mathbf{v}_i.$$

Connection: ρ_i^2 is an eigenvalue of $CC^\top$.

From Population CCA to Sample CCA

In practice, replace population covariance matrices by sample estimates:

$$\Sigma_{XX}, \Sigma_{YY}, \Sigma_{XY} \longrightarrow S_{XX}, S_{YY}, S_{XY}.$$

Compute sample canonical correlations $\hat{\rho}_1, \ldots, \hat{\rho}_r$ and coefficients.

Applied considerations

- Variables should be measured on the *same observations*.
- Covariance estimates must be sufficiently well conditioned.
- With many variables relative to n, sample correlations may be unstable or inflated.

Are the Canonical Correlations Nonzero?

A common global null hypothesis is

$$H_0 : \Sigma_{XY} = 0 \quad \Longleftrightarrow \quad \rho_1 = \cdots = \rho_r = 0.$$

Wilks' lambda is

$$\Lambda = \prod_{j=1}^{r} (1 - \hat{\rho}_j^2).$$

Under multivariate-normal sampling and standard regularity conditions, Bartlett's large-sample approximation is

$$-\left[n - 1 - \frac{p+q+1}{2}\right] \log \Lambda \,\dot{\sim}\, \chi^2_{pq}.$$

Interpretation: Small Λ indicates stronger evidence of association between the two variable sets.

Sales Example [**Source**: PSU STAT 505]

A firm surveyed a random sample of 50 employees to identify factors associated with sales performance. Two sets of variables were measured:

- Sales Performance: `Sales Growth`, `Sales Profitability`, `New Account Sales`$\Rightarrow p = 3$
- Intelligence Test Scores: `Creativity`, `Mechanical Reasoning`, `Abstract Reasoning`, `Mathematics`$\Rightarrow q = 4$

We will use canonical correlation analysis to explore the relationships between the two sets of variables.

Sales Example: Estimated Canonical Correlations

For $p = 3$ sales measures and $q = 4$ assessment scores, there are three canonical pairs.

```
cc1 <- cc(sales, intelligence)
cc1$cor
```

Pair	$\hat{\rho}_i$	$\hat{\rho}_i^2$
1	0.9945	0.9890
2	0.8781	0.7711
3	0.3836	0.1472

The first pair is very strongly correlated; the third is much weaker.

Caution: $\hat{\rho}_i^2$ describes the squared correlation between *two canonical variates*, not the fraction of total variance explained in the original variables.

Sales Example: Sequential Tests

```
rho <- cc(sales, intelligence)$cor
n <- dim(sales)[1]
p <- length(sales); q <- length(intelligence)
## Calculate p-values using the F-approximations
library(CCP)
p.asym(rho, n, p, q, tstat = "Wilks")
```

H_0	Approximate F value	p-value
$\rho_1^* = \rho_2^* = \rho_3^* = 0$	87.39	~ 0
$\rho_2^* = \rho_3^* = 0$	18.53	8.25×10^{-14}
$\rho_3^* = 0$	3.88	0.028

At the 5% level, the reported tests provide evidence that all three dimensions are nonzero $\Rightarrow$ all three canonical variate pairs may contribute to describing the relationship between the two sets of variables.

Obtain the Canonical Coefficients

	U_1	U_2	U_3
Growth	0.0624	-0.1741	-0.3772
Profit	0.0209	0.2422	0.1035
New	0.0783	-0.2383	0.3834

The first canonical variable for sales is

$$U_1 = 0.0624X_{\text{growth}} + 0.0209X_{\text{profit}} + 0.0783X_{\text{new}}$$

	V_1	V_2	V_3
Creativity	0.0697	-0.1924	0.2466
Mechanical	0.0307	0.2016	-0.1419
Abstract	0.08956	-0.4958	-0.2802
Math	0.0628	0.0683	-0.0113

The first canonical variable for test scores is

$$V_1 = 0.0697Y_{\text{create}} + 0.0307Y_{\text{mech}} + 0.0896Y_{\text{abstract}} + 0.0628Y_{\text{math}}$$

Canonical Coefficients: What Is Being Combined?

For the first sales variate, the reported coefficients give

$$U_1 = 0.0624X_{\text{growth}} + 0.0209X_{\text{profit}} + 0.0783X_{\text{new}}.$$

For assessment scores,

$$V_1 = 0.0697Y_{\text{creativity}} + 0.0307Y_{\text{mechanical}} + 0.0896Y_{\text{abstract}} + 0.0628Y_{\text{math}}.$$

Interpretation: These weights define the two composites that have the strongest sample correlation.

Caution: Raw coefficients depend on variable units and can be unstable when predictors within a set are correlated.

Canonical Loadings: What Do Variates Represent?

A canonical loading measures the correlation between an original variable and its set's canonical variate.

Sales variable	$\mathrm{Cor}(X_j, U_1)$	Score variable	$\mathrm{Cor}(Y_k, V_1)$
Growth	0.980	Creativity	0.638
Profit	0.946	Mechanical	0.721
New accounts	0.952	Abstract	0.647
		Mathematics	0.944

U_1 reflects broad sales performance; V_1 is strongly associated with mathematics and moderately with the other scores.

Cross-Loadings: Connecting the Two Sets

A **cross-loading** correlates an original variable with the *opposite* set's canonical variate.

Sales variable	$\mathrm{Cor}(X_j, V_1)$	Score variable	$\mathrm{Cor}(Y_k, U_1)$
Growth	0.975	Creativity	0.635
Profit	0.941	Mechanical	0.717
New accounts	0.947	Abstract	0.644
		Mathematics	0.939

For each pair, cross-loadings equal the corresponding own-set loadings multiplied by ρ_i (with a consistent sign convention).

How to Report the Sales Findings

An evidence-based summary

The strongest canonical dimension links a broad sales-performance composite with an assessment-score composite, especially mathematics. The estimated canonical correlation is 0.9945.

Do not conclude:

- That assessment scores *cause* better sales performance.
- That 98.9% of all sales variation is explained.
- That these results necessarily generalize beyond this sample.

The sample is small ($n = 50$) relative to the seven variables; validation is valuable.

Practical CCA Workflow

1. **Define** two scientifically meaningful sets of variables.
2. **Inspect** distributions, correlations, and missing data.
3. **Estimate** canonical correlations and variate coefficients.
4. **Assess** significance and practical strength of dimensions.
5. **Interpret** loadings and cross-loadings, not just weights.
6. **Validate** stability and communicate substantive findings.

Takeaways and Discussion

- CCA identifies maximally correlated linear composites of **two variable sets**.
- Canonical correlations measure association between *paired composites*.
- Coefficients construct variates; loadings help interpret them.

Discussion question

When would you use CCA rather than PCA, multiple regression, or a multivariate regression model?